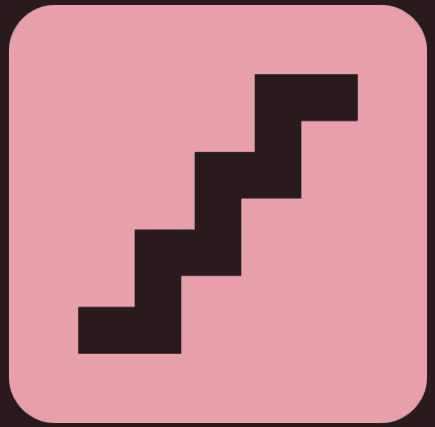
Higher National Diploma in Network Engineering, Year 2
BSc (Hons) Ethical Hacking & Network Security, Year 2

PRINCIPLES OF INFORMATION SECURITY

Synthesis and exam preparation



Day 10 - Afternoon

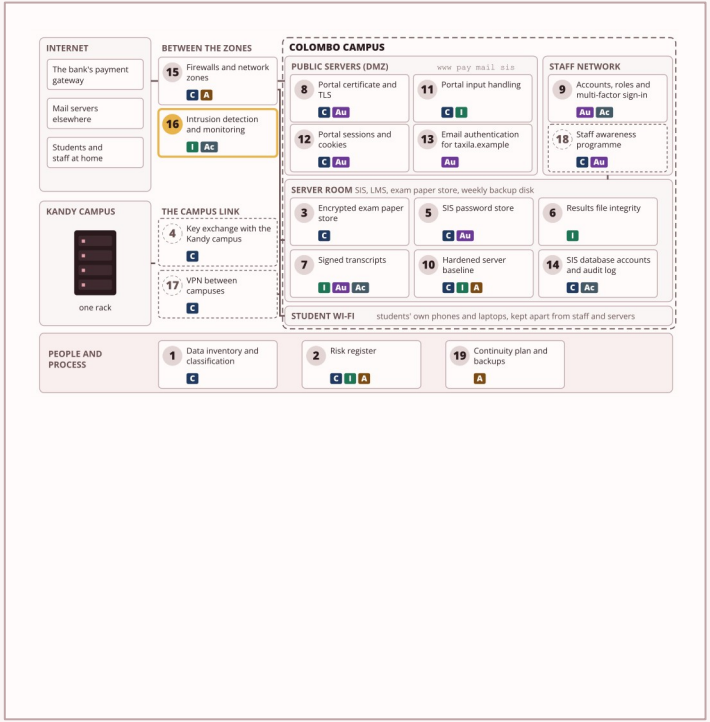
Running case: Taxila College, a fictional college

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Today

Walk back down the nine days, keep the two lists apart, then practise answering.

- The spine, end to end - 40 min
- The two lists, one more time - 30 min
- Answering the practice questions - 40 min



All six outcomes, one afternoon

LO1 Explain the fundamentals of information security.

LO2 Discuss the key characteristics of cryptography and cryptographic systems.

LO3 Analyze the principles of encryption, decryption, hashing, and data integrity.

LO4 Outline the process of ensuring digital security.

LO5 Describe the fundamentals of threats and countermeasures.

LO6 Explain the fundamentals and process of business continuity planning.

From this morning

Which item on Taxila's register helped your group most this morning, and which one failed you? Say which earlier idea it rests on.

CONCEPT 1

The spine, end to end

The nineteen register items, read in order, retell the whole placement.

40 min

Day 1: what we protect, and why

RECALL

Data and the five properties, from Day 1.

- Data has value, so it is an **asset**
- Five properties say what must hold for it
- **Risk** combines likelihood and cost
- Items 1 and 2: the data inventory and the risk register

Everything later rests on this floor.

Day 2: secrecy with keys

RECALL

Keys and the hybrid pattern, from Day 2.

- Encryption protects **confidentiality**
- A symmetric key is fast but hard to share
- A key pair solves the sharing
- The **hybrid** pattern: a key pair sets up a fast key
- Items 3 and 4: the exam paper store and the key exchange with Kandy

Rests on Day 1: it protects one property, confidentiality.

Day 3: fingerprints

RECALL

Hashing and salting, from Day 3.

- A hash turns any input into a short **digest**
- Any change gives a different digest
- A **salt** makes each stored digest unique
- Items 5 and 6: the password store and the results file check

Rests on Day 1's integrity, not on Day 2's keys: a hash has no key.

Day 4: proving who

RECALL

Signatures, certificates and transport security, from Day 4.

- A **signature** is a digest locked with a private key
- A certificate binds a name to a public key
- Secure web traffic uses both to prove the portal is genuine
- Items 7 and 8: signed transcripts and the portal certificate

Rests on two floors at once: Day 2's key pairs and Day 3's digests.

Day 5: who may do what

RECALL

The three processes set against the five properties, from Day 5.

- Authentication, authorisation, accounting: in that order
- **Least privilege**: only what the job needs
- A second approver for a released grade
- Items 9 and 10: accounts and roles, the hardened server

Answers the Registrar: can we prove who changed it?

Days 6 to 9: each part of the college

RECALL

Web input and sessions, email authentication and database least privilege, network zones, and attacks, risk and continuity, from Days 6 to 9.

- Day 6: input is never read as a command (items 11, 12)
- Day 7: email checks, scoped database accounts (items 13, 14)
- Day 8: zones, firewalls, monitoring, the Kandy link (items 15 to 17)
- Day 9: attacks, treatments, continuity (items 18, 19)

Each rests on Day 5's roles and Day 1's properties.

Climbing back down

Every control on the register answers a problem from Taxila's story.

YOU ALREADY KNOW THIS

You built a building one floor at a time, each floor on the one below. Now you climb back down, naming each floor as you pass.

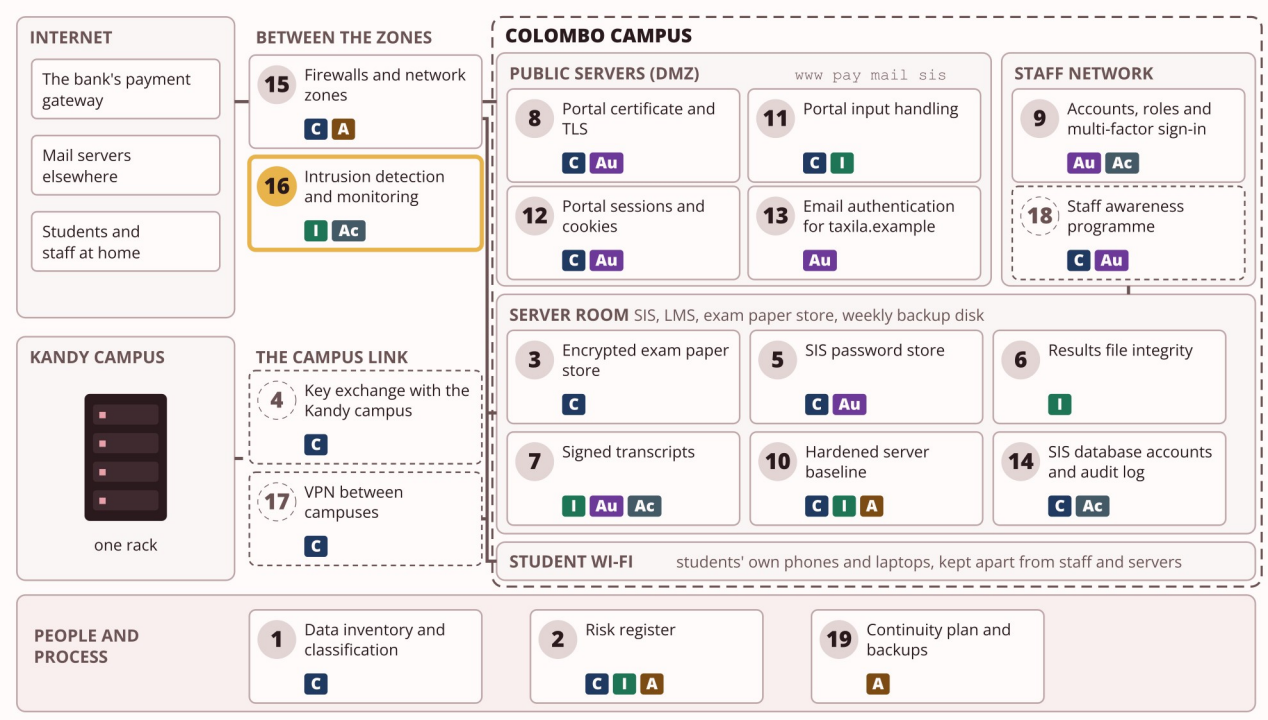
MAPPING

each floor you pass = one day's idea · **the ground you reach =** the five properties that everything rested on

T TAXILA

The nineteen register items, read in order, retell the whole placement.

The security register, complete



KEY

- Added earlier
- Demonstrated
- Concept only

CONCEPT 2

The two lists, one more time

The changed grade touched both: the integrity property and the accounting process.

30 min

The first list: what must hold

C CONFIDENTIALITY	I INTEGRITY	A AVAILABILITY	Au AUTHENTICITY	Ac ACCOUNTABILITY
Only the right people can read it.	Nobody has changed it without permission.	It works when it is needed.	It really comes from who it says.	You can prove who did what.
TAXILA *The leaked exam paper.*	TAXILA *The changed grade.*	TAXILA *The dead fee page.*	TAXILA *The fake Registrar email.*	TAXILA *Nobody knew who changed the grade.*

Five **properties**: each is something that must stay true.

Two lists that look alike

Five properties

What they say

What must hold for the data and the service.

The list

- Confidentiality, integrity, availability
- Authenticity, accountability

Three processes

What they say

How a system checks and records.

The list

- Authentication, authorisation
- Accounting

A process term is not a property term.

Words that look alike

- **Authentication** is the check at sign-in
- **Authenticity** is the property of being genuine
- **Accounting** is recording each action
- **Accountability** is being able to prove who did what

YOU ALREADY KNOW THIS

Two keys on your ring look almost the same. Use the wrong one and nothing opens.

MAPPING

the five-property key = what must hold · **the three-process key =** how a system checks and records

The changed grade, on both lists

- A grade changed without permission: **integrity** lost
- Nothing recorded who changed it: **accounting** missing
- So nobody could prove who did it: **accountability** lost

T TAXILA

The changed grade touched both: the integrity property and the accounting process.

× COMMON MISTAKE

The properties and the processes are one list.

They are two lists, and the right term is expected.

CONCEPT 3

Answering the practice questions

The nine practice questions used other Sri Lankan settings for exactly this reason.

40 min

What a practice question asks

- Each starts from a short scenario unlike Taxila
- So **recall alone** will not answer it
- Its parts rise in demand: explain, apply, then weigh
- Name mechanisms precisely and say **why**, not only what

YOU ALREADY KNOW THIS

A driving test takes you onto roads you have never practised on. You still pass, because you learnt to drive, not one route.

MAPPING

unfamiliar roads = an unfamiliar scenario · **the skills still transfer =** the ideas transfer beyond Taxila

Nine questions, nine settings

9
Practice questions
one per teaching day

3
Parts each
explain, apply, weigh

0
Set at Taxila
always somewhere new

Question	Setting
1	A district clinic's appointment system
2	A school examinations board
3	An online bookshop
4	A provincial land records office
5	A tea estate's payroll office
6	A doctor-channelling booking platform
7	A microfinance cooperative's mobile app
8	A courier company's tracking site
9	A ride-hailing app

Three parts, rising in demand

Explain

Say what it is and how it works

Name the mechanism, then the steps that make it work.

Not enough alone: a definition.

Apply

Use it on the setting

Map, design or find the flaw in the question's own case.

Use the setting, not Taxila.

Weigh

Judge a choice

Compare options; say what each gains and what it costs.

Say what could still go wrong.

Each part stands on the one before it.

Naming it, or explaining it

Only names

The bookshop's passwords

"They should use salting."

What it shows

The right word, and nothing more.

Explains

The bookshop's passwords

A unique random salt for each customer, stored with the row and added before hashing. Same passwords then give different digests, and a table built in advance matches none.

What it shows

What it is, where it lives, what it does, why it works.

Saying why a mechanism works earns the marks.

Timed practice across the topics

60 min

1. Answer one practice question per topic against the clock: three rounds of 15 minutes
2. Take one from each third: question 1 or 5; question 2, 3 or 4; question 6, 7, 8 or 9
3. Mark each other's answers for understanding, not wording
4. List the mechanisms you could name but not yet explain, and revise those



Marking a partner's answer

1. Does it name the right mechanism?
2. Does it say why the mechanism works?
3. Does it use the scenario, not Taxila?
4. Does the last part weigh a choice?

Mark understanding, not wording.

Common mistakes

× COMMON MISTAKE

Naming a mechanism is the same as explaining it.

Showing why a mechanism works earns the marks, not only its name.

× COMMON MISTAKE

The properties and the processes are one list.

They are two lists, and the right term is expected.

After today

READ	**Your notes for all nine days** And the nine practice questions, with their guidance
PRACTICE	**The module-wide ladder and quiz** On this day's page; they draw on all nine days
TASK	**Your group's after-action report** And the updated continuity plan

Questions & Discussion

Next: your group's after-action report, then revision from the spine.



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Revision 984f863